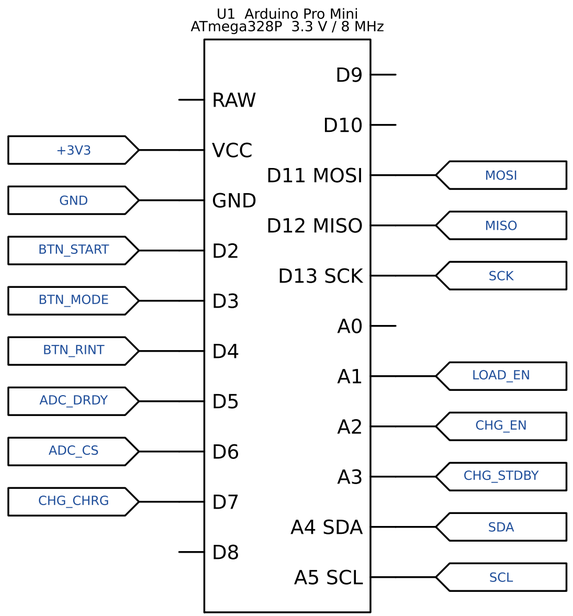
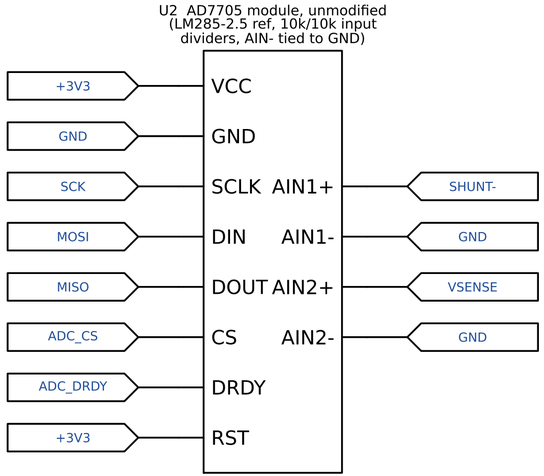
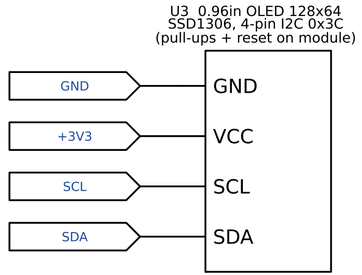
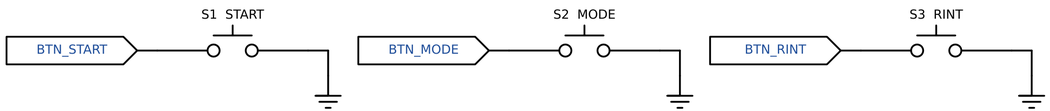






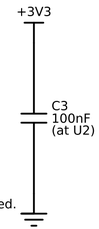


start test / back, hold = abort

QUICK -> FULL -> CHG+DIS

Rint test / back, hold = abort

internal pull-ups on D2, D3, D4

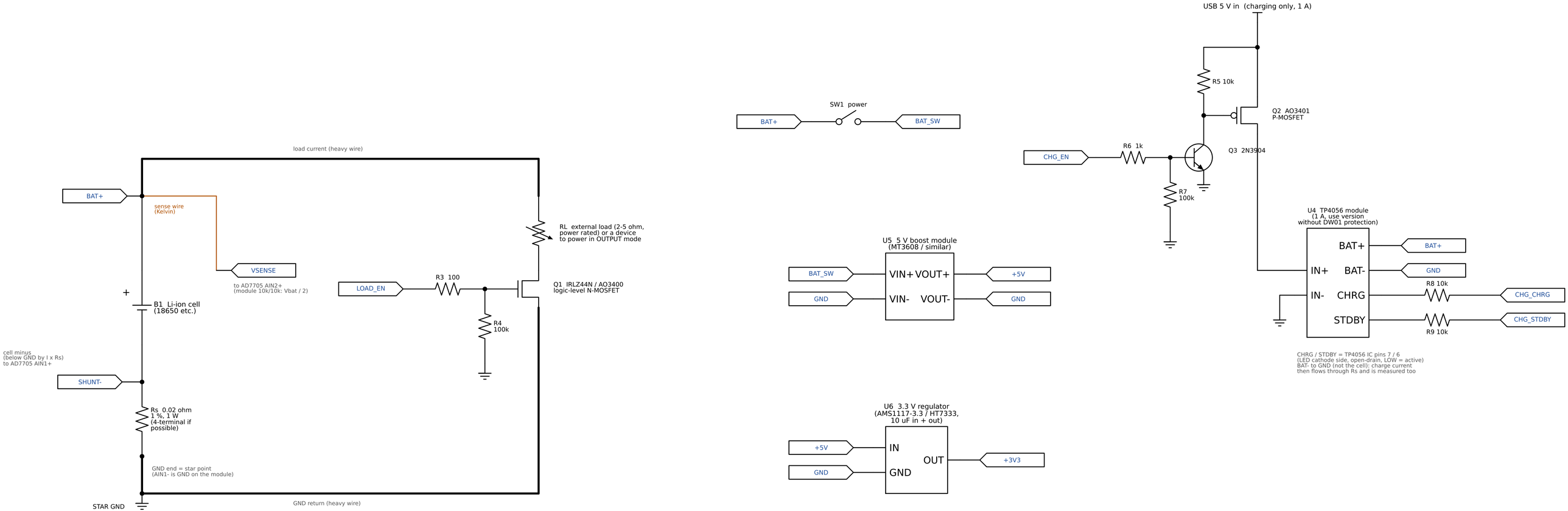


Notes
- +3V3 comes from U6 (sheet 2) into the Pro Mini VCC pin; RAW is unused.
- SPI (SCK/MOSI/MISO) is used by the AD7705 only. D8, D9, A0 unused; D10 must stay unconnected.
- AD7705 RST tied to +3V3; the firmware resets it over SPI.
- OLED on I2C (A4 SDA, A5 SCL); the 4-pin module has its own pull-ups and reset.
- GND = star point at the GND end of the shunt Rs (sheet 2).
- Program via the Pro Mini FTDI header (DTR, TX, RX, VCC, GND).
- Net labels with the same name are connected.

Li-ion Battery Analyzer

Sheet 1/2 - Controller, ADC, display, controls

Pro Mini 328 + AD7705 + 0.96in I2C OLED Rev F 2026-10-26



CHRG / STDBY = TP4056 IC pins 7 / 6
(LED cathode side, open-drain, LOW = active)
BAT- to GND (not the cell): charge current
then flows through Rs and is measured too

Notes
- Rs sits in the cell minus lead, so ALL cell current is measured: load + electronics (boost) + charging.
- STAR GND is the GND end of Rs. Q1 source, U5/U6, AD7705 module GND and TP4056 BAT- all join there.
- SHUNT- (cell minus) goes to AD7705 AIN1+ on its own thin wire; the module GND wire is the other sense lead.
- The AD7705 module's 10k/10k dividers halve both inputs: the voltage wire needs no external divider.
- Shunt channel: bipolar, gain 64, sign inverted in firmware; up to 2.5 A (-25 mV at the chip, limit -30 mV).
- The electronics draw ~30-60 mA from the cell. Switch SW1 off after a test, or the cell keeps draining.
- Programming: do not connect the FTDI adapter VCC while the analyzer runs from the cell.
- R4 / R7 keep the load and charger off while the MCU resets or is being programmed.
- No temperature sensing: do not leave high-current tests unattended.

Li-ion Battery Analyzer

Sheet 2/2 - Power, load, shunt, charger

Pro Mini 328 + AD7705 + 0.96in I2C OLED Rev F 2026-10-26